Polynomial-time simulation of non-Clifford quantum error correction

Arthur Pesah

FU Berlin

Joint work with



Serban Cercelescu

Oxford



Mark Koch

Oxford

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

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Most magic-state preparation protocols

Gates outside the Clifford group, such as T and CCZ.

These resources occur in many fault-tolerant protocols.

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

2 **Simulated**

The process of running the simulation on a classical computer.

This allows for the study of quantum systems without the need for actual quantum hardware.

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

3 **Polynomial time**

For the supported circuit class, the simulation cost grows polynomially with circuit size.

The claim is restricted to this class of circuits.

Phase polynomial states

Unitary gate set: the third-order CNOT-dihedral group

$$G_3 = \langle \text{CNOT}, T, X \rangle$$

Commuting T through the CNOTs: all diagonal non-Clifford gates of the third-level, i.e. T, CS, CCZ

Initial state: $|0\rangle^{\otimes n} |+\rangle^{\otimes m}$

Question: what happens when we apply G_3 gates to $|0\rangle^{\otimes n} |+\rangle^{\otimes m}$